

# Aditya Kulkarni

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**To the Amazon Robotics Engineering Team,**

Amazon Robotics is building the intelligent infrastructure that moves millions of packages to customers every day—at a scale where software correctness, system reliability, and AI-powered automation directly translate into real-world outcomes. The Software Development Engineer role sits at that intersection: cloud-native distributed systems, GenAI tooling, and software that interacts with physical fulfillment technology. I am an engineer who has spent the last year shipping production AI systems at scale, and I am excited to bring that to Amazon Robotics.

- **Distributed Systems at Scale:** At **Isomer AI**, I owned the production backend—PostgreSQL, Redis, Docker, Terraform—managing CI/CD, resolving on-call incidents, and maintaining SLA across enterprise environments. I also built **CortexQ**, a Kubernetes-native LLM inference platform with KEDA autoscaling, multi-provider model serving, and full OpenTelemetry observability, load tested to 500 concurrent users.
- **GenAI & AI-Powered Development:** I deployed 10+ production AI features using multi-model pipelines spanning Claude, GPT-4o, and Gemini via MCP tooling and Temporal.io agentic workflows—923 commits, 7 deployments. I built an **LLM Eval Harness** tracking hallucination rate, latency, and cost across 2,000 labeled examples with Bootstrap CIs and McNemar significance tests. Leveraging GenAI for development productivity is already how I work.
- **Cloud-Native Architecture on AWS:** Production experience across AWS and GCP: automated CI/CD deployments, Grafana monitoring, Kafka real-time streaming pipelines, and Helm/ArgoCD GitOps. AWS is my primary cloud environment.
- **Research-Grade Software Engineering:** As a **Graduate Research Assistant at NC State**, I built cloud infrastructure serving 500GB+ of genomic data to 100+ researchers and improved ML model accuracy by 12% via feature engineering on biological sensor data—instilling a standard of correctness and operational reliability that maps directly to fulfillment-center software.

Amazon Robotics operates at a scale where software quality has physical consequences. That standard of engineering is what I want to be held to. I would be glad to discuss how my experience in production AI systems and cloud-native infrastructure translates to your team's roadmap.

Sincerely,

Aditya Kulkarni